# Vital signs ruler — children (acceptable ranges by age)

paediatric · term neonate – 17 years · casualty / ward

**Paeds vitals by age**

## ACCEPTABLE RANGES BY AGE

| Age | ≈ kg | Heart rate /min | Resp rate /min | Systolic BP mmHg |
|---|---|---|---|---|
| | | 40 75 110 145 180 | 10 24 38 51 65 | 50 72 95 118 140 |
| **Term** | 3.5 | **120–170** | **25–60** | **60–95** |
| **3 mo** | 6 | **115–170** | **25–60** | **60–105** |
| **6 mo** | 8 | **110–170** | **20–55** | **75–105** |
| **1 y** | 10 | **105–150** | **20–45** | **70–105** |
| **2 y** | 12 | **95–150** | **20–40** | **70–105** |
| **4 y** | 15 | **80–150** | **17–30** | **75–110** |
| **6 y** | 20 | **75–140** | **16–30** | **80–115** |
| **8 y** | 25 | **70–130** | **16–30** | **80–115** |
| **10 y** | 30 | **60–130** | **15–25** | **85–120** |
| **12 y** | 40 | **65–120** | **15–25** | **90–120** |
| **14 y** | 50 | **60–115** | **14–25** | **90–125** |
| **16 y** | 60 | **60–115** | **14–25** | **90–130** |
| **17 y +** | 65 | **60–115** | **14–25** | **90–135** |

## PEARLS & PITFALLS

Ranges are 'acceptable' for a calm child; crying/fever raise HR & RR

Use the right cuff: bladder ≈ 40 % of arm circumference

## FORMULA

### Bar = acceptable range for age

- **Weight estimate (1–10 y)**: (age + 4) × 2 kg
- **Hypotension (1–10 y)**: SBP < 70 + 2 × age mmHg
- **Hypotension (< 1 y)**: SBP < 70 (term neonate < 60) mmHg

## RELATED (SA)

Tachypnoea: < 60 d > 60 · 2–12 mo > 50 · 1–5 y > 40 · 5–12 y > 25 /min ✚

Oxygen if $SpO_2$ < 92 % ✚

Compensated shock: cold peripheries, weak pulses, CRT ≥ 3 s, ↑ HR, altered LOC — hypotension is late and ominous ✚

SATS triage: younger child < 3 y or < 95 cm; older child 3–12 y or 95–150 cm

**RCH** Royal Children's Hospital Melbourne CPG — Acceptable ranges for physiological variables, updated Jul 2023 · **SATS** Twomey M et al. Vital signs for children at triage: revised South African Triage Scale (SATS) for children. S Afr Med J 2013;103(5):304-8 · **APLS** APLS / PALS bedside formulae (weight estimate, minimum SBP) — classic teaching · **CLASSIC** classic teaching · ✚ **PAED** STG & EML Paediatric Hospital Level 5th ed. (updated Mar 2026) — p. 1.15, 15.2, 15.5

✚ = SA STG / national guideline (pages in footer). Unmarked lines: the other sources listed. Checked against PocketDoc\Guidelines 2026-10-31. **Card** vitals-paeds · updated 2026-10-31 · ANDH district level